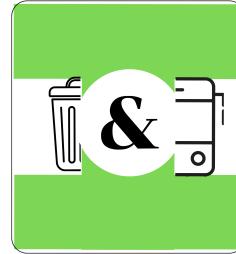


First Draft

REDINEST - TEAM BLUE



THROW & TRAVEL
SMART-BIN & SUSTAINABLE MOBILITY NETWORK

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Stakeholder Analysis

Investors and Interests

Waste collection companies or agencies

They will be able to access data to improve the management efficiency and reporting system. One possible result could be the improvement of the number of recycling points if the average of the recycling process increases with time.

Sustainability Mobility partners

In Hungary, MOLBubi or Lime can invest with us to increase the usage of sustainable branding. Any other developing company can partner with us to reach more environmentally conscious users.

If the solution does not reach action, adopting sustainable habits by citizens will decrease with generational change. Improving recycling rates will take more time and cost. Finally, there will be a loss of a customer-based channel, which can be a good step towards the field of sustainability and digital transformation.

| Core idea – need/problem, target group, competitors, stakeholders

1. A case title:

- Throw and Travel

Note: We can use recycling and travel to avoid “throw” as it sounds unprofessional for a project.

2. Problem:

- Insufficient citizen participation in proper waste disposal and recycling.
- A lack of accurate real-time visibility into waste and recycling volumes across the city. (*Mohu Mol waste management in Hungary*) *France does not have the system.*

Note: We can conduct a quick research to see any numbers or methods of collecting such data in the city or around the world in general to back up our claims.

- Inefficient recycling initiatives for positive behaviour.

3. Solution:

Our smart-bin platform makes recycling engaging through sustainable travel rewards while providing municipalities with real-time waste data.

3.1. Main Impacts

- Through gamification, waste disposal will be more accessible for younger generations.
- Accurate tracking and monitoring of recycled waste through AI reporting will be organized and developed.
- Sustainable and environmentally friendly transportation will be supported by rewarding recycling behaviour through an incentive point system.
- Community feedback mechanisms strengthen participation in improving green solutions.

4. Basic Concepts

- Desirability
- Feasibility
- Viability
- The triple bottom line
- Define customers and consumers

5. Competitors

- City system/waste management
- Deposit Return Systems (*Like “Volta” in Portugal, i.e.*)

6. Selling Value

How much are we going to sell it for? How are we going to finance ourselves?

7. Our Advantage

- Free to use;
- Giving people something in return for recycling waste;
- Engaging people;
- Remove deposit return value on legible items;

8. Empathy Canvas?

8.1. Persona

- Young person (student)
- Reka
- Hungarian
- 22 years

8.2. Say and Do?

- She says that there is a way to motivate people to recycle

8.3. See?

- People are not doing recycling properly

8.4. Pains

- Not finding a proper way to throw trash

8.5. Gains

- Earn money
- Contribute to the city's recycling

8.6. Hear

- It's difficult to achieve what she wants, to become financially stable/comfortable.

8.7. Think and feel

- Want to save money
- Believes in sustainability
- Feels that she has a mission as a member of society

8.8. What is your solution? (*One Sentence*)

- We help Citizens get rewarded and contribute to the city's recycling through our smart-bin platform, and contribute to sustainable travel rewards while providing municipalities with real-time waste data.

8.8.1. Desired Change & Impact Measurement

- As a result of our solution, we want citizens to be more motivated to recycle and actively engage in sustainable measures. (*Social Change*)

8.8.2. Impact Measurement

- Number of active users on the platform
- Behaviour change rate
- Avoided waste (Kg)

non profit organization

supported by the government

Individual contribution

9. Presentation Checklist

9.1. Core Idea: *need/problem, target group, competitors, stakeholders*

- Insufficient citizen participation in proper waste disposal and recycling.
- A lack of accurate real-time visibility into waste and recycling volumes across the city. (Mohu Mol waste management in Hungary) France does not have the system.
- Note: We can conduct a quick research to see any numbers or methods of collecting such data in the city or around the world in general to back up our claims.
- Inefficient recycling initiatives for positive behaviour.
-
- Our business idea is a citizen-based recycling project. We aim to reduce street pollution and encourage citizens, especially the younger generation, to recycle by using our application. The app is based on gamification, allowing users to earn rewards when they recycle correctly. These reward points can be exchanged for free public transportation or discounts on government-supported services, such as electric vehicle charging stations or museum visits. We have identified some similar existing systems as indirect competitors, but currently we do not have any direct competitors. Our main stakeholder is the government, and in the future, we hope to attract private companies as additional partners. Through this project, we want to create cleaner cities while making recycling more engaging and beneficial for citizens.

9.2. Digital Transformation: *The role and significance of digital transformation*

Our solution transforms traditional waste management into a smart digital ecosystem by combining QR codes, artificial intelligence, cloud technology, and a mobile application. Instead of simply throwing away recyclable waste, users scan a QR code on the smart bin, allowing the system to securely identify them and record their recycling activity. AI then verifies the type of waste and checks whether it has been disposed of correctly. Once confirmed, users automatically receive reward points that can be redeemed for public transportation, bike sharing, e-scooter rides, or even museum and theater tickets. At the same time, every recycling activity is stored in the cloud, providing municipalities with real-time data on recycling rates, waste composition, and bin usage. This data enables smarter decision-making, more efficient waste collection, lower operational costs, and supports the development of more sustainable cities

9.3. Green & Sustainability: *Green and sustainability aspects*

9.3.1. Green

Motivating people into recycling with gamification and rewards, improving the participation of the citizens in recycling. Raising more awareness.

Real-time tracing in the waste system, improving feasibility in garbage collecting, using people's data to analyze and improve recycling operations.

Improving public transportation use with our rewards system.

9.3.2. Sustainability

Transforming recycling operations to a more sustainable form by increasing public involvement and traceability.

Creating a sustainable economical environment with government founding and sponsorships with private companies.

9.4. Economic & Competition: *Economic and competition aspects*

Companies providing smart bins and waste collection optimization

exist. For example, Sensoneo is a global leader in data-driven smart waste management and operational efficiency. Our idea takes this one step further to introduce gamification and community feedback to strengthen and support sustainable travel options.

9.5. Data: *Making digital data understandable (what data you rely on and how and why?)*

9.5.1. What Data Do We Rely On?

We track two key data points:

- User Behaviour: who is recycling, when, and where;
- Bin Data: which includes live fill levels, material types, and contamination rates.

9.5.2. How Do We Collect It?

The user scans a QR code on our app to link their profile. Inside the bin, IoT sensors check its capacity, while an AI optical scanner instantly verifies the exact type of waste being deposited.

9.5.3. Why We Rely On It?

This data makes our entire ecosystem viable. For the citizen, AI verification prevents fraud, allowing us to instantly convert their good habits into travel rewards. For waste managers, it eliminates blind spots. Instead of driving fixed routes to empty half-full bins, cities use our live dashboard to dispatch trucks only where needed, saving fuel, cutting emissions and giving sponsors concrete analytics on their environmental impact.

9.6. Business Models: *Potential revenue sources and business models*

9.6.1. Revenue sources

- Government - The public sector is our primary partner. Because increasing recycling rates and gaining accurate, real-time data on waste flows is a clear public interest, the government finances the operation and maintenance of the platform through a fixed contribution. In return, the city receives a reliable tool for tracking, reporting and improving its waste management, ensuring the long-term stability of the service.
- Private companies - Our second source of revenue comes from the partner companies whose rewards are offered through the platform. For every reward redeemed by a user, the company pays us a commission, as our platform brings them new, engaged customers. For example, if a user collects 1,000 points with a value of 10€, we retain a 10% commission, which the partner company covers in exchange for the acquired customer and the visibility gained within the application.

9.6.2. Plans for the future

As the platform grows, two further revenue streams can be introduced. The first is a share of the value generated by the recycling process itself, earned through the sale of the collected materials to recycling operators. The second is advertising space within the application, offered to sustainability-oriented brands. Together, these would diversify our income and reduce dependence on any single source.

9.6.3. Business model

Key Partners • Municipalities & government • Recycling operators • Transport & mobility partners • Technology providers (IoT, AI)	Key Activities • Operating smart bins (IoT & AI sorting) • Running the rewards app • Collecting & reporting waste data	Value Propositions • Citizens: free recycling rewarded with travel benefits • Cities: real-time waste data & less pollution	Customer Relationships • Gamified, self-service app • Points & incentives • Community feedback	Customer Segments • Citizens (mainly youth/students) • Municipalities • Recycling operators • Sponsor companies
	Key Resources • Smart-bin hardware & sensors • AI verification software • Recycling & user data		Channels • Mobile app & QR codes • Smart bins in public spaces • Partner locations	

Cost Structure

- Smart-bin production & maintenance
- Software & data infrastructure
- Reward funding & operations

Revenue Streams

- Government: pays a fixed amount to run and maintain the platform
- Private companies: commission on redeemed rewards (e.g. a user collects 1,000 points worth 10€ → we keep 10%)
- Future plans: a share from recycling (sale of collected materials) and in-app advertising

9.7. Inclusion & Urban Development: *Inclusion, sensitive issues and urban development*

Including the younger generation into recycling. Choosing smart bins locations carefully (placing them to be reachable for as citizens we can. Designing bins in order to make it easily usable for kids and people with disabilities.

9.8. Connect All the Dots

Visual metaphor of data

the bar charts with the solution impact on the amount of trash

one if the bars will be visualized with trash and the man on one of them

it will decrease with implementation

Final Scripts:

Throw & Travel – 5-Minute Pitch Script

Slide 1 – Introduction (40 seconds)

Good morning everyone. We are Team Blue, and today we'd like to introduce **Throw & Travel**, a smart recycling platform that rewards people for making sustainable choices.

Research shows that people, especially Gen Z, are more likely to repeat a behavior when they receive immediate feedback and meaningful rewards. This is why reward-based apps like Duolingo, fitness trackers, and loyalty programs are so successful.

Unfortunately, recycling doesn't work this way. People recycle because they know it's the right thing to do, but they receive no recognition or immediate benefit. As a result, participation remains lower than it could be.

Our idea is simple: **Recycle. Earn. Travel.** By combining AI, QR codes, smart bins, cloud technology, and a mobile app, we transform recycling into a rewarding digital experience.

Slide 2 – Core Idea (45 seconds)

Imagine you're walking home after finishing a drink.

Instead of simply throwing the bottle into a recycling bin, you scan the QR code on our smart bin using the Throw & Travel app.

The system securely identifies you, and an AI camera automatically recognizes whether you've disposed of the correct recyclable material.

Within seconds, you receive reward points on your phone.

Those points can later be exchanged for public transport tickets, bike-sharing, electric scooter rides, or even museum and theater tickets.

At the same time, the city receives valuable recycling data that traditional recycling bins cannot provide.

Everyone benefits—the citizen, the city, and the environment.

Slide 3 – Digital Transformation (50 seconds)

Our platform demonstrates digital transformation by replacing a traditional recycling process with a smart, connected digital ecosystem.

QR codes securely connect recycling activities to registered users.

Artificial intelligence verifies waste automatically, reducing contamination and improving recycling quality.

Cloud computing stores every recycling activity in real time, while the mobile app provides instant rewards and user engagement.

Instead of relying on manual monitoring, municipalities gain real-time insights into recycling rates, waste composition, and bin fill levels.

This enables smarter waste collection, lower operational costs, and better environmental decision-making.

Slide 4 – Green & Sustainability (35 seconds)

Our solution creates environmental benefits in several ways.

First, gamification motivates more people to recycle consistently.

Second, real-time monitoring allows waste collection trucks to optimize their routes, reducing unnecessary fuel consumption and carbon emissions.

Finally, by rewarding sustainable transportation instead of cash, we encourage environmentally friendly lifestyles while supporting long-term sustainable cities.

Slide 5 – Economy & Competition (35 seconds)

There are already companies providing smart bins and waste monitoring systems.

However, our competitive advantage is that we focus on changing human behavior.

Rather than simply collecting waste, we encourage people to participate through rewards, community engagement, and sustainable mobility.

This creates value not only for municipalities but also for citizens and private partners.

Slide 6 – Making Data Understandable (40 seconds)

Data is at the heart of our platform.

We collect information about recycling frequency, waste type, contamination levels, and smart-bin status.

The QR code securely links recycling activities to users, while AI verifies every disposal.

This information allows municipalities to optimize waste collection schedules, predict when bins need to be emptied, reduce fuel costs, and lower greenhouse gas emissions.

Better data leads to better decisions.

Slide 7 – Revenue & Business Model (35 seconds)

Our platform is free for citizens.

Revenue comes from organizations that benefit from improved recycling.

Municipal governments pay a service fee because the platform improves recycling efficiency and reduces operational costs.

Private companies can sponsor rewards to increase their visibility and support sustainability initiatives.

In the future, we can also generate revenue through recycled-material partnerships, premium analytics, and environmentally focused advertising.

Slide 8 – Inclusion & Urban Development (30 seconds)

Throw & Travel is designed for everyone.

Smart bins are placed in accessible public locations and designed to be easy to use for children, older adults, and people with disabilities.

By making recycling simple, engaging, and rewarding, we encourage participation across the entire community and help build cleaner, more sustainable cities.

Slide 9 – Conclusion (45 seconds)

To conclude, Throw & Travel is more than a smart recycling bin.

It is a digital platform that changes recycling behavior.

By combining AI, QR technology, cloud computing, and gamification, we motivate citizens to recycle while giving municipalities the real-time data they need to make smarter decisions.

Citizens receive meaningful rewards.

Cities become cleaner and more efficient.

Businesses gain opportunities to participate in sustainable initiatives.

Our vision is simple:

Don't just ask people to recycle—give them a reason to keep doing it.

Together, we can build cleaner cities, smarter communities, and a more sustainable future.

Thank you for listening

Tab 2

For a project like this, the financial model should ideally combine **public-sector funding, private-sector partnerships, and data-service revenues**, so that the platform does not depend on only one source of income.

Cost Structure

Initial Development Costs

- Smart bin hardware (sensors, QR/NFC scanners, cameras if AI is used)
- Mobile application development
- AI waste classification and reporting system
- Cloud infrastructure and database
- Pilot testing and installation

Operational Costs

- Platform maintenance and software updates
 - Server and cloud hosting
 - Customer support
 - Reward management (travel credits, discounts, vouchers)
 - Marketing and community engagement
 - Data analytics and reporting
-

Revenue Sources

1. Municipality Subscription (Main Revenue Source)

Cities pay an annual or monthly fee for access to:

- Real-time waste monitoring dashboard
- Recycling statistics and reporting
- Predictive analytics for waste collection routes
- Community participation metrics

Example:

- Small city: €10,000–20,000/year
- Medium city: €30,000–60,000/year
- Large city: €100,000+/year

This is similar to a Software-as-a-Service (SaaS) model.

2. Smart Bin Licensing

Municipalities purchase or lease the smart bins.

Revenue can come from:

- Direct sale of bins
- Monthly leasing fees
- Maintenance contracts

Example:

- Smart bin lease: €20–50/month per bin
-

3. Public Transportation Partnerships

Since users earn sustainable mobility rewards:

Potential partners:

- MÁV Group
- BKK
- Local bike-sharing systems

The municipality or transport company subsidizes reward points because increased public transport use supports sustainability goals.

4. Corporate Sponsorships

Companies can sponsor green challenges and reward campaigns.

Examples:

- Supermarkets
- Coffee chains
- Retail stores
- Sustainable brands

Example:

"Recycle 20 bottles this month and receive a discount voucher."

Companies pay:

- Sponsorship fees
- Advertising placements inside the app
- Campaign participation fees

5. ESG and Sustainability Reporting Services

Many organizations need environmental reporting.

The platform can provide:

- Recycling impact reports
- Carbon reduction estimates
- Community engagement metrics

Potential clients:

- Universities
 - Shopping centers
 - Corporate campuses
 - Industrial parks
-

6. Grants and EU Funding

This type of project aligns strongly with:

- European Commission Green Deal initiatives
- Smart Cities programs
- Circular Economy Action Plan
- Horizon Europe innovation funding
- Erasmus+ sustainability projects

These can fund pilot implementation and scaling.

7. Data Analytics Services

Aggregated and anonymized data can help municipalities understand:

- Which districts recycle most
- Peak disposal times
- Waste composition trends
- Collection route optimization

Cities pay for advanced analytics modules.

Example Business Model (Budapest Pilot)

Expenses

- App development: €40,000
- AI reporting system: €25,000
- 50 smart bins: €50,000
- Marketing and rewards: €15,000

Total initial investment: ~€130,000

Annual Revenue

- Municipality subscription: €50,000
- Smart bin leasing: €24,000
- Corporate sponsorships: €20,000
- Data analytics services: €15,000

Total annual revenue: ~€109,000

This means the project could approach operational break-even within 1–2 years if adoption targets are met.

Strongest Value Proposition for Investors

The most attractive aspect is that the project solves **three problems simultaneously**:

1. Increases recycling participation through gamification.
2. Provides municipalities with real-time waste intelligence.
3. Encourages sustainable transportation through reward incentives.

For a startup competition or funding pitch, I would summarize the revenue model as:

B2G (Business-to-Government) SaaS platform for smart waste management, complemented by smart-bin leasing, ESG reporting services, corporate sponsorships, and sustainable mobility partnerships.

This gives the project recurring revenue rather than relying only on hardware sales.

Final Outcome

Core Idea

Our business idea is a citizen-based recycling project. We aim to reduce street pollution and encourage citizens to recycle by allowing users to earn rewards when they recycle correctly through our app. These reward points can be exchanged for free public transportation or discounts on government-supported services.

Digital Transformation: *The role and significance of digital transformation*

QR code marked bin with an AI camera to identify the type of waste (plastic, paper, metal, glass, etc.). Once the AI confirms the correct disposal, users automatically receive reward points in the mobile app. These points can be redeemed for sustainable transportation services such as public transit, bike sharing, or electric scooter rides and other services like museum tickets. Every recycling transaction is stored in a cloud database, allowing municipalities to monitor recycling rates.

Green & Sustainability: *Green and sustainability aspects*

Motivating people into recycling with gamification and rewards, improving the participation of the citizens in recycling. Raising more awareness.

Real-time tracing in the waste system, improving feasibility in garbage collecting, using people's data to analyze and improve recycling operations.

Sustainability

Transforming recycling operations to a more sustainable form by increasing public involvement and traceability.

Creating a sustainable economical environment with government founding and sponsorships with private companies.

Economic & Competition: *Economic and competition aspects*

Companies providing smart bins and waste collection optimization exist. For example, Sensoneo is a global leader in data-driven smart waste management and operational efficiency. Our idea takes this one step further to introduce gamification and community feedback to strengthen and support sustainable travel options.

1.1.1. What Data Do We Rely On?

- User Behaviour: who is recycling, when, and where;
- Bin Data: which includes live fill levels, material types, and contamination rates.

1.1.2. Why We Rely On It?

This data makes our entire ecosystem viable. For the citizen, AI verification prevents fraud, allowing us to instantly convert their good habits into travel rewards. For waste managers, it eliminates blind spots. Instead of driving fixed routes to empty half-full bins, cities use our live dashboard to dispatch trucks only where needed, saving fuel, cutting emissions and giving sponsors concrete analytics on their environmental impact.

Revenue sources

- Government - The public sector is our primary partner. Because increasing recycling rates and gaining accurate, real-time data on waste flows is a clear public interest, the government finances the operation and maintenance of the platform through a fixed contribution. In return, the city receives a reliable tool for tracking, reporting and improving its waste management, ensuring the long-term stability of the service.
- Private companies - Our second source of revenue comes from the partner companies whose rewards are offered through the platform. For every reward redeemed by a user, the company pays us a commission, as our platform brings them new, engaged customers. For example, if a user collects 1,000 points with a value of 10€, we retain a 10% commission, which the partner company covers in exchange for the acquired customer and the visibility gained within the application.

Including the younger generation into recycling. Choosing smart bins locations carefully (placing them to be reachable for as citizens we can. Designing bins in order to make it easily usable for kids and people with disabilities.

Professionally done in the ppt

Inclusion in the pitch

Consuming a lot of energy to make something green

Calculate and make a credible claim about something

Statistics about our claim

Credit of taxes these points of redeeming system

Go to market strategy our idea is going to be an add on

The added value the updated service with technology model behaviour

Modelling behaviour through

So why would you change the whole system if they already get money and have a travel pass

Or include the food stamps of the supermarkets

More streamline pipeline to have the burden taken away because they already have it

One-page document

1. Core Idea: *need/problem*

- Insufficient citizen participation in proper waste disposal and recycling.
- Inefficient recycling initiatives for positive behaviour.

We will reduce street pollution and encourage citizens, especially the younger generation, to recycle by using our application. The app is based on gamification, allowing users to earn rewards when they recycle correctly. These reward points can be exchanged for free public transportation or discounts on government-supported services, such as electric vehicle charging stations or museum visits.

2. Digital Transformation: *The role and significance of digital transformation*

There is a need for an integration of a smart digital ecosystem to combine QR codes, artificial intelligence, cloud technology, and a mobile application. Instead of simply throwing away recyclable waste, users scan a QR code on the smart bin, allowing the system to securely identify them and record their recycling activity. AI then verifies the type of waste and checks whether it has been disposed of correctly. Once confirmed, users automatically receive reward points that can be redeemed for public transportation, bike sharing, e-scooter rides, or even museum and theater tickets. At the same time, every recycling activity is stored in the cloud, providing municipalities with real-time data on recycling rates, waste composition, and bin usage. This data enables smarter decision-making, more efficient waste collection, lower operational costs, and supports the development of more sustainable cities.

3. Green & Sustainability: *Green and sustainability aspects*

Motivation is one factor affecting people to participate in good behaviour. One way to increase it is with gamification and reward, raising more awareness for sustainability. Real-time tracing in the waste system, improving feasibility in garbage collecting, using people's data to analyze and improve recycling operations.

4. Economic & Competition: *Economic and competition aspects*

Companies providing smart bins and waste collection optimization exist. For example, Sensoneo is a global leader in data-driven smart waste management and operational efficiency. Our idea takes this one step further to introduce gamification and community feedback to strengthen and support sustainable travel options.

5. Data: *Making digital data understandable*

We track two key data points:

- User Behaviour: who is recycling, when, and where;
- Bin Data: which includes live fill levels, material types, and contamination rates.

The user scans a QR code on our app to link their profile. Inside the bin, IoT sensors check its capacity, while an AI optical scanner instantly verifies the exact type of waste being deposited.

This data makes our entire ecosystem viable. For the citizens, AI verification prevents fraud, allowing us to instantly convert their good habits into travel rewards. For waste managers, it eliminates blind spots. Instead of driving fixed routes to empty half-full bins, cities use our live dashboard to dispatch trucks only where needed, saving fuel, cutting emissions and giving sponsors concrete analytics on their environmental impact.

6. Business Models: *Potential revenue sources and business models*

Revenue sources

The public sector is our primary partner. Because increasing recycling rates and gaining accurate, real-time data on waste flows is a clear public interest, the government finances the operation and maintenance of the platform through a fixed contribution. In return, the city receives a reliable tool for tracking, reporting and improving its waste management, ensuring the long-term stability of the service.

Our second source of revenue comes from the partner companies whose rewards are offered through the platform. For every reward redeemed by a user, the company pays us a commission, as our platform brings them new, engaged customers. For example, if a user collects 1,000 points with a value of 10€, we retain a 10% commission, which the partner company covers in exchange for the acquired customer and the visibility gained within the application.

Plans for the future

As the platform grows, two further revenue streams can be introduced. The first is a share of the value generated by the recycling process itself, earned through the sale of the collected materials to recycling operators. The second is advertising space within the application, offered to sustainability-oriented brands. Together, these would diversify our income and reduce dependence on any single source.

Business model

Key Partners • Municipalities & government • Recycling operators • Transport & mobility partners • Technology providers (IoT, AI)	Key Activities • Operating smart bins (IoT & AI sorting) • Running the rewards app • Collecting & reporting waste data	Value Propositions • Citizens: free recycling rewarded with travel benefits • Cities: real-time waste data & less pollution	Customer Relationships • Gamified, self-service app • Points & incentives • Community feedback	Customer Segments • Citizens (mainly youth/students) • Municipalities • Recycling operators • Sponsor companies
	Key Resources • Smart-bin hardware & sensors • AI verification software • Recycling & user data		Channels • Mobile app & QR codes • Smart bins in public spaces • Partner locations	
Cost Structure • Smart-bin production & maintenance • Software & data infrastructure • Reward funding & operations		Revenue Streams • Government: pays a fixed amount to run and maintain the platform • Private companies: commission on redeemed rewards (e.g. a user collects 1,000 points worth 10€ → we keep 10%) • Future plans: a share from recycling (sale of collected materials) and in-app advertising		

Tab 5

To visualize our solution we can see today's situation with high waste output. According to very recent studies, although people care about the environment, recycling participation is still limited.

With Throw & Travel we have an inverse relationship where engagement will be increasing to reach the sustainable target which is going the opposite direction.

And this is the beginning of change. Combining technology with behavioral incentives, we can create cleaner cities and smarter waste management.